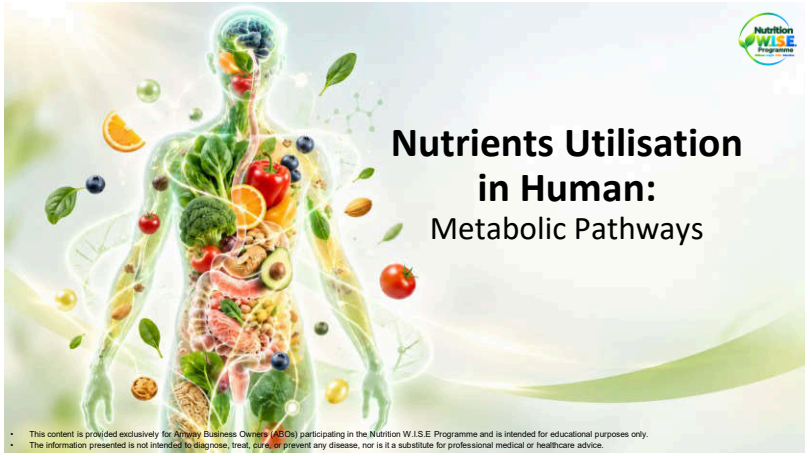


Nutrients Utilisation in Human:
Metabolic Pathways

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1

Food – Source of Nutrients and Energy

Can you identify?

1. Carbohydrates
2. Protein
3. Fats



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2

Food – Source of Nutrients and Energy

1. **Carbohydrates**
   - ✓ Rice, cucumber, sugar in sambal
2. **Protein**
   - ✓ Hard-boiled egg, anchovies, peanuts
3. **Fats**
   - ✓ Coconut milk, cooking oil



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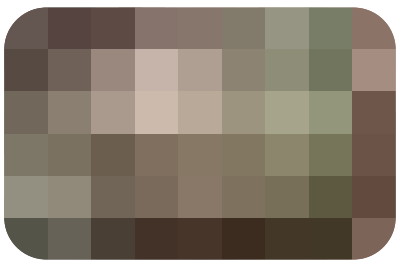
Measuring Energy: Calories in Your Food

A calorie (kcal) is simply a unit of energy.
Every gram of carbohydrate, protein, and fat releases a different amount when your body burns it.

Carbohydrates → e.g. Rice	**4 kcal per gram**
Protein → e.g. Chicken	**4 kcal per gram**
Fats → e.g. Cooking oil	**9 kcal per gram**

How many calories is the food?

Total estimate: roughly 550–650 kcal



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4

Where Your Energy Goes?

BMR, DIT & Activity



Basal Metabolic Rate (BMR)

About 60 – 70%

Energy used just to keep your body running — breathing, heartbeat, brain, keeping warm — even lying in bed all day.



Diet-induced Thermogenesis (DIT)

About 5 – 15%

Energy your body spends digesting, absorbing, and processing the rice, chicken, and oil you just ate.



Physical Activity (ExEE and NEPA)

About 10 – 20%

- **ExEE** - running, swimming, cycling, badminton etc
- **NEPA** - standing, talking, gardening, cooking etc

Energy expenditure varies with age, muscle mass, and activity level

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5

Energy Balance: Calories In vs Calories Out



Source:
Manoboli, J.M., Maglios, F. Compensatory Responses to Exercise Training As Barriers to Weight Loss: Changes in Energy Intake and Non-exercise Physical Activity. Curr Nutr Rep 12, 327–337 (2023)

Resting metabolic rate (RMR), diet-induced thermogenesis (DIT), and physical activity energy expenditure (PAEE) that can be further divided to exercise energy expenditure (ExEE) and non-exercise physical activity (NEPA).

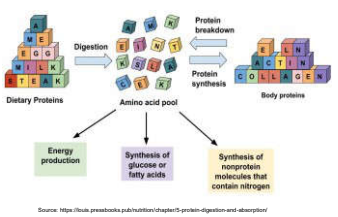
Weight gain occurs when energy balance is positive, and weight loss occurs when energy balance is negative.

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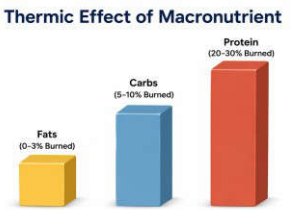
6

Protein Has The Highest Thermic Effect

Protein produces more thermic effect, followed by carbohydrates and fats.



Source: https://louis.pressbooks.pub/nutrition/chapter/5-protein-digestion-and-absorption/



Protein produces more thermogenesis (heat production) than fat and carbs because your body uses a lot of energy just to digest, absorb, and process it.

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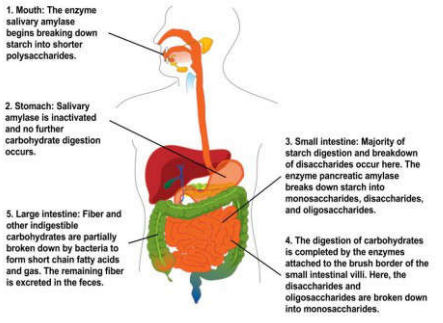
Carbohydrate Metabolism

Nutrients Utilisation in Human: Metabolic Pathways

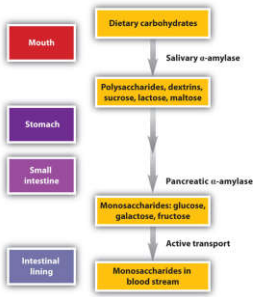
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From Rice to Glucose: Digestion & Absorption



https://med.libretexts.org/Courses/Chabot_College/Introduction_to_Nutrition_Science/04:_Carbohydrates/4.02:_Digestion_and_Absorption_of_Carbohydrates



https://step1.medbullets.com/gastrointestinal/110027/carbohydrate-digestion

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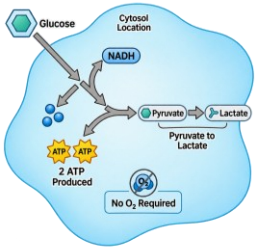
Glycolysis:
Turning Glucose into Energy

Inside each cell, glucose from your rice is taken apart step by step to release usable energy — the first and fastest way your body taps into carbohydrate.

Glucose — 1 sugar molecule, inside the cell

Glycolysis — a chain of about 10 small chemical steps breaks it apart

Pyruvate + Energy — 2 smaller molecules, plus usable energy (ATP)

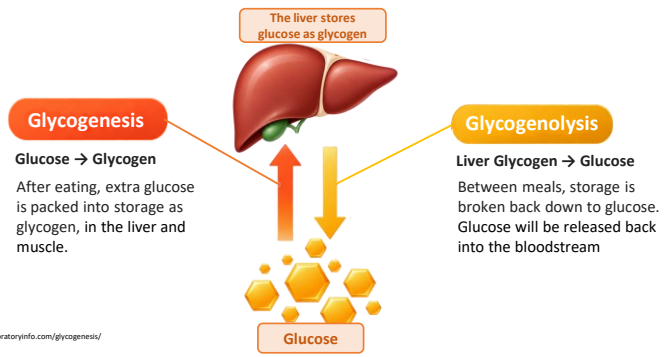


Source: https://metabolomics.creative-proteomics.com/resource/glycolysis-vs-oxidative-phosphorylation-atp.html

Glycolysis happens in every cell, doesn't need oxygen, and is a rapid way your body produces energy from glucose.

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Storing & Releasing Sugar

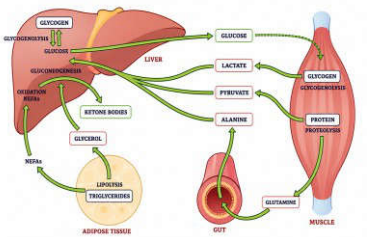


Source: https://laboratoryinfo.com/glycogenesis/

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Gluconeogenesis:
Making New Glucose

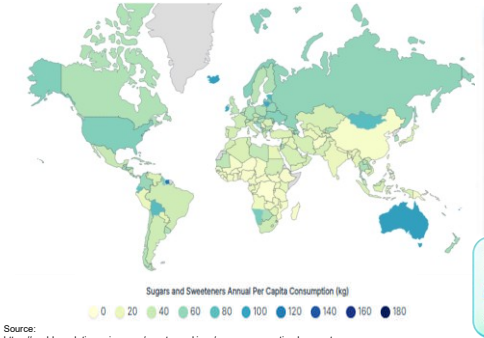
1. **How does your body maintain blood glucose levels when you're not eating?** Your body produces new glucose from non-carbohydrate sources through a process called gluconeogenesis.
2. Protein breakdown, Fat breakdown — **Main Building Blocks:** Amino acids (from protein)/ Glycerol (from fat)
3. **The Liver rebuilds these leftover pieces** into new glucose
4. **New Glucose is released** into blood, keeps your brain fueled



This mainly happens during fasting, sleep, or long exercise — it's your body's backup glucose supply, protecting your brain, which relies heavily on it.

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Which country has the highest sugar consumption per capita



COUNTRY	SUGARS AND SWEETENERS ANNUAL PER CAPITA CONSUMPTION 2022 (KG)
Luxembourg	168.15
Fiji	138.70
Suriname	125.35
Montenegro	112.32
Australia	102.58
Iceland	102.31
Lithuania	98.84
Ireland	96.73
Nauru	93.88
Macau	84.67



Source:
https://worldpopulationreview.com/country-rankings/sugar-consumption-by-country

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Lipid Metabolism

Nutrients Utilisation in Human: Metabolic Pathways

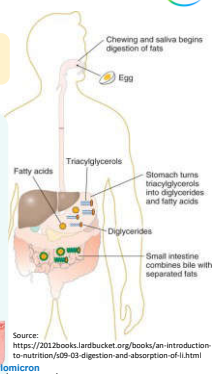
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From Oil to Fatty Acids:
Digestion & Absorption

Fat doesn't dissolve in water, so your body needs a special process to break it apart and carry it away

Dietary Fat	Stomach	Bile	Pancreatic Lipase	Micelles	Absorption
Oil used in cooking your meal	Fat starts to break into smaller droplets as it mixes with food. A small amount is also digested by gastric lipase.	Made in the liver and stored in the gallbladder. Works like dish soap on a greasy pan, breaking large fat droplets into smaller ones so enzymes can work more effectively.	Breaks down each triglyceride into 2 free fatty acids and 1 monoglyceride	Bile salts wrap around the fatty acids and monoglycerides, forming tiny carriers that shuttle them through the watery gut fluid to the intestinal cell surface.	Fatty acids & monoglycerides enter intestinal cells, where most are rebuilt into triglycerides and packed into chylomicrons → lymph → bloodstream



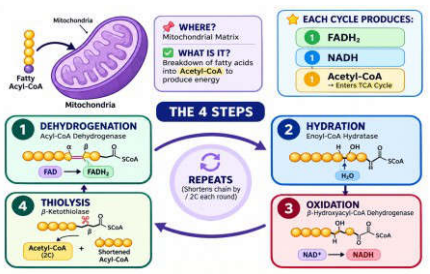
Source:
https://2012books.lardbucket.org/books/an-introduction-to-nutrition/s09-03-digestion-and-absorption-of-l-l.html

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Beta-oxidation:
Breaking Down Fat for Energy

- **A fatty acid is a long chain** your body clips it apart, two carbons at a time, to release energy
- **Fat provides slower but sustained energy** and because it is stored abundantly, is especially important during fasting, rest, and prolonged low-to-moderate activity.
- **Fatty Acid** — Long carbon chain from dietary fat or body fat stores
- **Beta-Oxidation** — repeatedly clipped into 2-carbon units
- **Acetyl-CoA feeds into the body's central energy cycle** and is used for ketogenesis



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Ketogenesis:
Backup Fuel from Fat

Ketones are a normal, healthy backup fuel
This is how the body keeps the brain fueled when glucose runs low

Happens when fasting, very low-carb eating, or prolonged exercise → carbs run low

Adipose tissue releases fatty acids and travel to the liver.

Liver breaks down fatty acids via β-oxidation
Produces acetyl-CoA

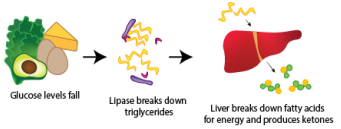
Acetyl-CoA is converted into ketone bodies
Mainly acetoacetate & beta-hydroxybutyrate

Ketones released into blood
Used as fuel by the brain, muscle, and heart

Typical Metabolism: Carbohydrates used for energy



Ketosis: Fat used for energy



Source:
https://my.clevelandclinic.org/health/articles/24660-ketosis-definition-symptoms.html

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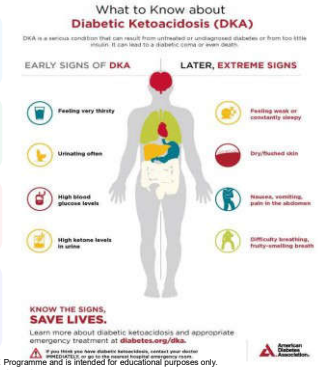
Ketones As Fuel & Consequences

Triggered by stress, starvation or diabetes
Stress, extreme starvation or diabetes can trigger the liver to convert fat into ketones for energy

Usually mild and reversible
In healthy people with normal insulin function, this is generally mild and reversible

Ketoacidosis is dangerous
Uncontrolled build-up of ketones that makes the blood too acidic

Medical emergency
Symptoms include excessive thirst, nausea, fruity breath, confusion, rapid breathing — requires urgent insulin and fluid treatment



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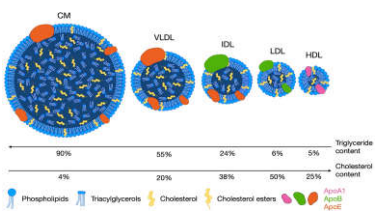
18

Fat Storage & Transport

Storage
Adipose Tissue
Extra fat from your meal is packaged into triglycerides and stored in fat cells (adipose tissue) as your body's long-term energy reserve

Transport
Lipoproteins

Chylomicrons	Carry dietary (diet-derived) triglycerides from the intestine to tissues after a meal
VLDL Very-low-density Lipoprotein	Carries triglycerides made by the liver out to tissues for energy or storage
LDL Low Density Lipoprotein	Transports cholesterol away from the liver, toward tissues throughout the body; if in excess, it can deposit cholesterol in artery walls
HDL High Density Lipoprotein	Mainly transports cholesterol back toward the liver for processing/excretion



Source:
Pérez-Medina, C., Fisher, E.A., Fayad, Z.A. et al. Radiolabeling lipoproteins to study and manage disease. Eur J Nucl Med Mol Imaging 52, 4717–4734 (2025).

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Protein Metabolism

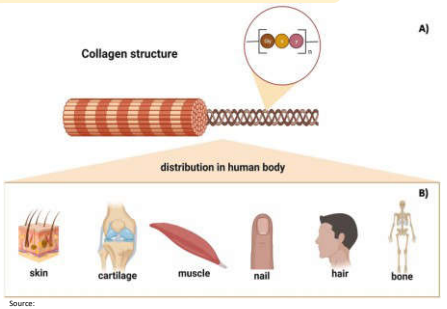
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Did You Know?
Collagen is the most abundant protein in the body

Collagen is basically your most abundant protein.
It makes up about **25-35%** of all the protein in your body, forming the scaffolding for skin, tendons, ligaments, and bones. It's also why bone broth and gelatin get so much hype, though eating collagen doesn't mean it goes straight to your skin.

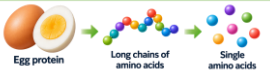


Source:
Campos, Luana Dias et al. "Collagen supplementation in skin and orthopedic diseases: A review of the literature." Heliyon vol. 9,4 e14961. 28 Mar. 2023, doi:10.1016/j.heliyon.2023.e14961

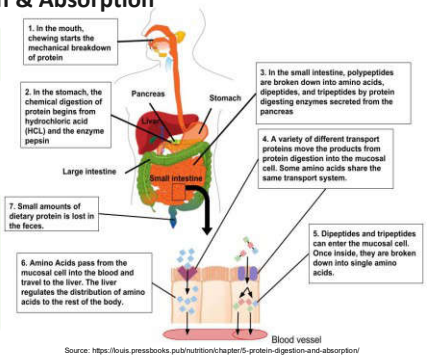
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From Egg to Amino Acids:
Digestion & Absorption

Protein is a long chain of amino acids linked together. Digestion breaks that chain apart, link by link.



- **Egg (Protein)** — long chains of amino acids
- **Stomach** — stomach acid + pepsin start breaking chains apart
- **Small Intestine** — enzymes finish the job: chains → single amino acids
- **Bloodstream** — amino acids are absorbed and travel to your cells



Source: https://louis.pressbooks.pub/nutrition/chapter/5-protein-digestion-and-absorption/

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The Amino Acid Pool

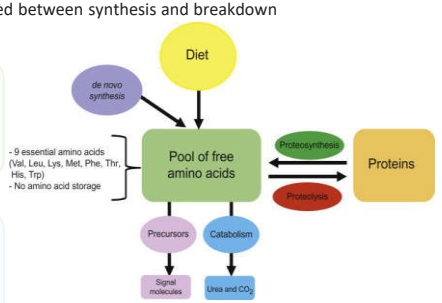
It is a constantly balanced between synthesis and breakdown

01 Inflows

- From digestion — amino acids from chicken you just ate
- From body protein — recycled from breakdown of old cells & tissue

02 Outflows

- Protein synthesis — builds new muscle, enzymes, hormones
- Energy / glucose — excess amino acids can be broken down for energy



Source:
Averous, J., Jousse, C., Maurin, A.-C., Bruhat, A., & Fafournoux, P. (2016). Chapter 20 - Adaptation to Amino Acid Availability: Role of GCN2 in the Regulation of Physiological Functions and in Pathological Disorders. In The Molecular Nutrition of Amino Acids and Proteins (pp. 249–260). Academic Press.

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Protein Synthesis:
Building New Proteins

Your body reads its own genetic instructions to link amino acids into brand-new proteins, built for a specific job

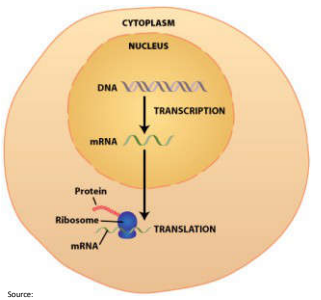
DNA in the nucleus holds the genetic code, the instructions for making each protein

Transcription — DNA is copied into mRNA (messenger RNA) inside the nucleus, which carries the genetic instructions out to the cytoplasm

mRNA travels to the ribosome — once in the cytoplasm, mRNA attaches to a ribosome, the cell's protein-building machinery

Translation — the ribosome reads the mRNA sequence and links amino acids together in the correct order to build a new protein chain

New Protein can be made, such as muscle protein actin and myosin



Source:
https://sites.duke.edu/apep/module-3-alcohol-cell-suicide-and-the-adolescent-brain/biology-and-chemistry-connections/dna-transcription-translation-synthesis-of-proteins/

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Bringing It All Together

Nutrients Utilisation in Human: Metabolic Pathways

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Bringing It All Together:
Integration of Metabolism

All three macronutrients break down into common intermediates

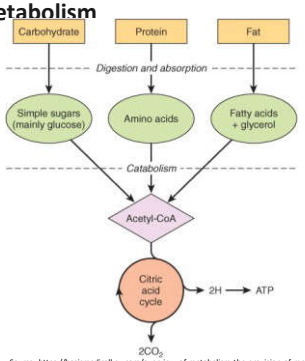
- Carbohydrates (via glucose)
- Fats (via fatty acids)
- Protein (via amino acids)

are each broken down through their own pathways, but converge on shared molecules like acetyl-CoA

Acetyl-CoA is the central meeting point

The **citric acid cycle (Krebs cycle)** is the shared energy hub

This shared pathway allows metabolic flexibility.
The body can switch between burning carbs, fat, or protein for energy depending on availability and demand.



Source: https://basicmedicalkey.com/overview-of-metabolism-the-provision-of-metabolic-fuels/

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Energy Balance

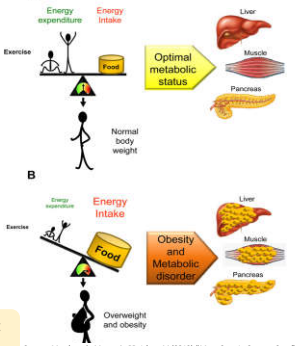
Fat cells grow in two ways:



The number of fat cells increases mostly during late childhood and early puberty

- After that, fat cell number stays fairly fixed — adults mainly gain fat by cells getting bigger, not more numerous
- This makes childhood weight management especially important, since early excess weight can permanently raise fat cell count

Simple tools like reading food labels can help prevent excess weight gain during these key growth years.



Source: Martínez de Morentin PB, López M (2010) "*Mens Sana In Corpore Sana*": Exercise and Hypothalamic ER Stress. PLoS Biol 8(8): e1000464.

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Fasting & Starvation :
What Happens When Food Runs Out

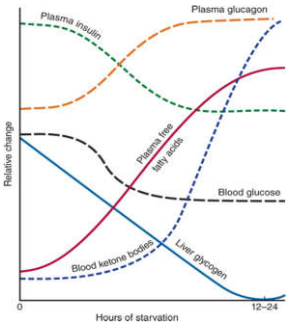
A few hours (between meals) — as blood glucose starts to dip, the body taps stored glycogen to keep levels stable

Overnight / skipped meal (8–24 hrs) — glycogen stores run low, so the liver starts gluconeogenesis

Prolonged fasting (1–3+ days) — fat breakdown speeds up, and the liver produces ketone bodies to fuel the brain and spare glucose

The body relies heavily on fat and ketones to conserve muscle, but as fat stores become severely depleted, protein breakdown rises sharply for energy and glucose

Starvation (extended, severe) — with fat reserves low, muscle breakdown becomes a major energy source, a sign of the body's last-resort survival response



Source: Karimi, R., Cleven, A., Elbarbry, F. et al. The Impact of Fasting on Major Metabolic Pathways of Macronutrients and Pharmacokinetics Steps of Drugs . Eur J Drug Metab Pharmacokinet 46, 25–39 (2021).

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Key Takeaways



Food = Fuel

Rice, egg, and oil are broken into glucose, amino acids, and fatty acids — each metabolized differently, but all convertible into energy



Balance Decides The Outcome

Whether energy is stored or burned comes down to calories in vs. calories out — and your body constantly manages that balance



The Body Is Adaptable

Carb, fat, and protein pathways are connected. When one fuel runs low, your body smoothly shifts to another — fed or fasting

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